

CHAPTER

1

Macroeconomics for the long run and for the short run

What is the subject matter of macroeconomics? What methods and simplifying assumptions do macroeconomists use in their efforts to explain how the economy works? These are the issues addressed in this introductory chapter.

The chapter explains the philosophy underlying our course in macroeconomics. In particular, it explains why we have divided the main body of the text into two Books, one concerned with the long run, and another one dealing with the short run. Reading this chapter will help you to understand the links between the following chapters and to see where the material in each individual chapter fits into the big scheme. In addition, this chapter presents some important concepts and elements of macroeconomic models which we will use repeatedly in later chapters.

We start by discussing how to define macroeconomics. We then go on to explain why it is useful to develop separate macroeconomic theories for the long run and for the short run, and we present the essential assumptions underlying these two branches of macroeconomics as well as their main implications. In particular, we explain how the assumptions behind macroeconomics for the long run lead to the idea of a 'natural' level of economic activity. We also provide a theoretical foundation for the existence of nominal wage and price rigidities which are crucial for understanding fluctuations in economic activity in the short run.

1.1 What is macroeconomics?

It seems natural for a macroeconomic textbook to start by giving a general definition of macroeconomics. Yet you are not going to see a perfectly clear definition here; we simply do not think that there is one. In fact, economists have even found it hard to offer a clear-cut definition of the general science of economics.

The economist and Nobel Prize winner Gary Becker once said that 'economics is the study of the allocation of scarce resources to satisfy competing ends'.¹ This certainly says something essential about what (mainly micro) economics is. However, some parts of macroeconomics are concerned with situations where resources are *not* scarce, because the available supplies of labour and capital are not fully utilized. These important real-world situations would not fall within the realm of economics according to the above definition.

Given the difficulties of providing a brief and accurate definition of economics, one should not be surprised that a perfectly clear subdivision into micro- and macroeconomics

1. The idea behind this definition goes back to the nineteenth century, and the first scholar to give a comprehensive statement of it was Lionel Robbins in his famous *Essay on the Nature and Significance of Economic Science*, London, Macmillan, First edition, 1932.

is also elusive. It is sometimes said that macroeconomics is that part of economics which is concerned with the economy as a whole. This suggests that microeconomics focuses only on the small elements of the economy such as the single agent or the market for a particular product. Although much of macroeconomics is concerned with the economy in the large, this distinction between macro and micro is inaccurate. Important parts of macroeconomics are not (directly) concerned with the whole economy, but rather with understanding particular markets such as the labour market or the credit market. And a large and important part of microeconomics, general equilibrium theory, is concerned with the *interaction* among markets, that is, with the economy as a whole.

We therefore think that the best characterization of macroeconomics is one that simply states the main questions asked in this branch of economics.

A definition of macroeconomics by subject

What creates growth in aggregate output and income per capita in the long run? And what causes the fluctuations in economic activity that we observe in the short run? These are the basic questions in macroeconomics. At the risk of oversimplifying, we may therefore say that *macroeconomics is the study of economic growth and business cycles*. As we shall see later on, to explain the movements in total output we must also understand the movements in total consumption, investment and the rate of unemployment, as well as the interaction of these real variables with nominal variables such as the general level of wages, prices, nominal interest rates, foreign exchange rates, etc. Hence macroeconomics also includes the study of these variables.

A definition of macroeconomics by method

What we have offered above is a definition of macroeconomics by subject: macroeconomics is defined by the issues studied by macroeconomists. A strict 'empiricist' version of this definition, which also involves the choice of method, is to say that *macroeconomics is concerned with explaining observed time series for economic variables like GDP, consumption, investment, prices and wages, the rate of unemployment, etc.* This reflects the view that a scientific discipline should be defined in terms of the data it seeks to explain.

We do not agree completely with this view. We do think there are some purely theoretical scientific contributions that should be considered as part of the body of macroeconomics. However, like the rest of the macroeconomics profession, we are heavily influenced by the empiricist view. To secure the link between theory and the real world, theories should be evaluated by holding them up against the facts, and new theory should ideally be justified and accompanied by illustrative empirical material. You will see that our book very much reflects this approach.²

Why do macroeconomists aggregate?

The variables entering into macroeconomic models are typically *aggregate* variables covering the economy as a whole. For instance, in macroeconomics we often describe the entire production side of the economy as if a single commodity were produced by the use of just two different inputs, capital and labour, both one-dimensional variables which can be represented by a number. By contrast, microeconomics studies disaggregated models in which it is typically not 'allowed' to aggregate the production of, say, oranges and apples into the production of fruit.

The aggregation undertaken in macroeconomic models raises obvious problems. Take the concept of the aggregate capital stock, for example. Capital is defined as produced means

2. According to this view it is important that one is familiar with the definitions of the main statistical series on macroeconomic variables. We do not include the definitions in this book, since we assume that you know the basic concepts of national income accounting and the standard measures of inflation and unemployment, etc., from an earlier course in macroeconomics.

of production, so it includes, for example, buildings as well as computers. How should the quantities of these two capital goods be added into one number representing their combined productivity? In practice, the aggregate real capital stock is measured by multiplying the quantities of the different capital goods by their respective prices in some base year, and then adding up the values of the stocks of buildings, computers, etc., calculated at the fixed base-year prices. This would seem to be a sensible way of measuring the quantity of aggregate capital input provided the relative prices of the different capital goods remain reasonably constant over time. But we know that over the past decades the relative price of computers has decreased enormously, and at the same time computers have become tremendously more productive. Using relative prices to obtain an aggregate measure of capital representing the productivity of produced inputs is therefore a dubious procedure.

Nevertheless, the assumption that capital input as well as total output can be represented by single numbers is standard in macroeconomics. How can we defend, for instance, letting the production of myriads of different goods and services be represented by one number called 'aggregate output'? There are several lines of defence:

1. Over time the outputs of a lot of goods and services – including capital goods – do in fact tend to move in the same direction. Given that the production volumes of most industries tend to be positively correlated, it seems defensible to use concepts like aggregate output or aggregate investment, even if we do not have a method of measuring these aggregates which is fully correct in all circumstances.
2. The economy is such a complex mechanism that we cannot hope to explain and describe it in all detail. To understand at least some of the economic regularities observed, we have to make strong simplifications by abstracting from many details. Aggregation of variables is one convenient way of simplifying.
3. Whether a certain highly simplifying assumption is useful or not is ultimately an empirical question. If a model built on strong simplifications yields predictions which accord with observed movements in some economic variables, then that model seems useful for understanding (some of) the determinants of those variables. Presumably, it will then also be useful for evaluating the effects of economic policies intended to affect the variables considered.

Economist Robert Solow (of whom you will hear a lot more in this course) has put it the following way:

All theory depends on assumptions which are not quite true. That is what makes it theory. The art of successful theorizing is to make the inevitable simplifying assumptions in such a way that the final results are not very sensitive.³

The long run versus the short run

From our discussion it should be no surprise that in our view the best definition of macroeconomics is:

MACROECONOMICS: THE STUDY OF GROWTH AND BUSINESS CYCLES



Macroeconomics is the part of economics seeking answers to the questions: 'what creates growth in income per capita in the long run?' and 'what creates fluctuations in aggregate income in the short run?' It also tries to answer some related questions such as: 'what explains the level of long-run unemployment?', and 'what explains the short-run variations in unemployment?'

3. Robert M. Solow, 'A Contribution to the Theory of Economic Growth', *Quarterly Journal of Economics*, 70, 1956, pp. 65–94.

Because these questions relate to different time horizons, we have organized the main body of this text in two Books, where the first is concerned with the long-run questions above, and the second is concerned with the short-run questions.

We find both parts of macroeconomics very important, but one can argue that the issues addressed in long-run macroeconomics are the most important ones. Consider a poor country like Uganda whose GDP per capita is only about 3 per cent of GDP per capita in the USA. A typical policy issue in long-run macroeconomics is: how could a country like Uganda initiate a growth process that would gradually take it up to, say, 20 per cent of the US level? A typical policy issue in short-run macroeconomics is: what could a government in, say, the UK, Sweden, Denmark or the Netherlands do to avoid an increase in the rate of unemployment from 5 to 8 per cent which would otherwise follow a negative shock to the economy? We do find the latter question very important, but for anyone concerned with the long-run welfare of human beings, the first type of question seems more essential.

1.2 Long-run and short-run macroeconomic phenomena

The distinction between long-run and short-run macroeconomics is first and fundamentally a distinction between the *phenomena we want to understand*. This leads to a distinction between the *fundamental characteristics of the models* we use, and of the *policies we analyse*. Below we will motivate these statements and explain the nature of the difference between long-run and short-run macroeconomic models.

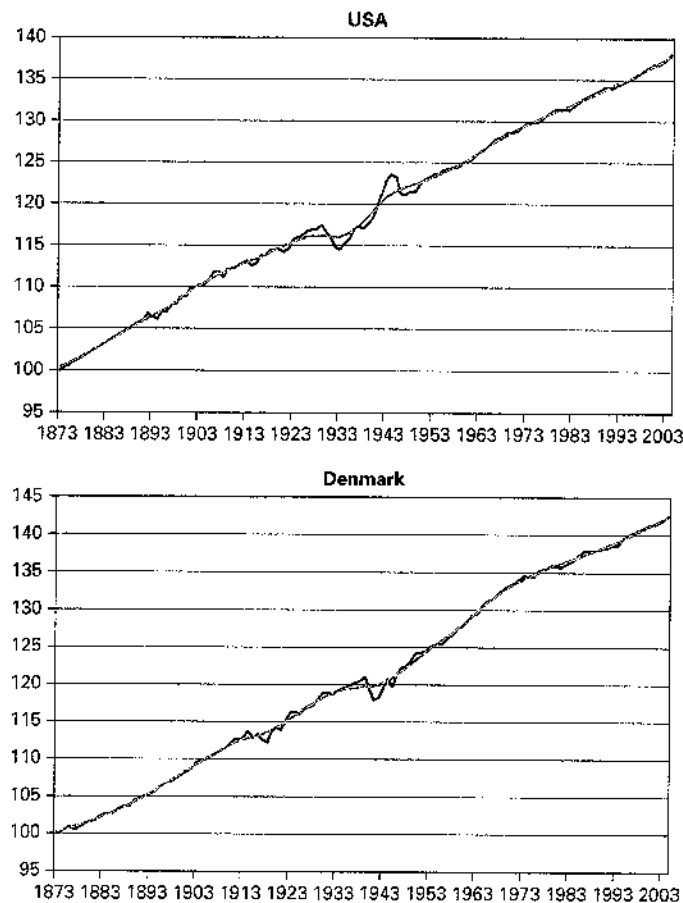
Let us start by looking at some data. In Figs 1.1 and 1.2 we have chosen to focus on two countries, the USA and Denmark, since these two nations can be seen as polar cases: the USA is a large and relatively closed economy with a fairly small public sector (by European standards), whereas Denmark is a small and relatively open economy with a large public sector. We want to illustrate that, despite these differences, some important qualitative features of the data are shared between the two economies. Similar figures drawn for countries like the UK, Sweden, the Netherlands, and for practically any developed market economy would show the same qualitative features.

In Fig. 1.1 the natural logarithms of the annual real GDPs of the USA and Denmark are drawn up for the period from 1873 to recently, with the value in 1873 indexed to 100 for both countries. Figure 1.2 shows the average annual rates of unemployment for the two countries during the past and present century. In both figures the actual data are represented by the most solidly drawn zig-zagged curves.

The figures also include grey curves which are much more smooth. These are meant to express the *trends* of the relevant series. In Chapter 13 you will learn about a technique for constructing such a trend (that technique has in fact been used here). For now, let us just view the trend curves intuitively as the curves which a man in the street would draw with a pencil if he were asked to illustrate the ‘underlying movement’ or the ‘underlying gravity level’.

In any event, the figures suggest that one way to interpret the movements of each of the series is to view them as being made up of two components: a *trend component* representing the overall evolution and captured by the smooth curves, and a *cyclical component* representing the year-to-year fluctuations and captured by the shifting vertical deviations between the actual data curve and the trend curve.

You may find that the cyclical components in Fig. 1.1 do not seem that great. To highlight the fluctuations (around the long-run trend), we have drawn Fig. 1.3 which shows the annual rate of change in real GDP, and the annual absolute change in the rate of unemployment for the USA and Denmark for the shorter period 1980–2008. The annual rate of change in GDP varies from sometimes higher than 6 per cent, down to sometimes around –2 per cent, and the rate of unemployment occasionally increases by up to two percentage points in just one year, and also sometimes drops by two points in one year. These cyclical movements are of considerable size. Final data for 2009 were not available at the time the manuscript for this book was completed, but the worldwide financial and economic crisis that exploded in the

FIGURE 1.1 Logarithm of real GDP in the USA and Denmark, 1873–2004

Note: Logarithm of GDP in 1873 normalized to 100. Trends constructed by HP-filtering with lambda set to 100.

Source: Angus Maddison, *The World Economy: Historical Statistics*, and Penn World Table 6.2.

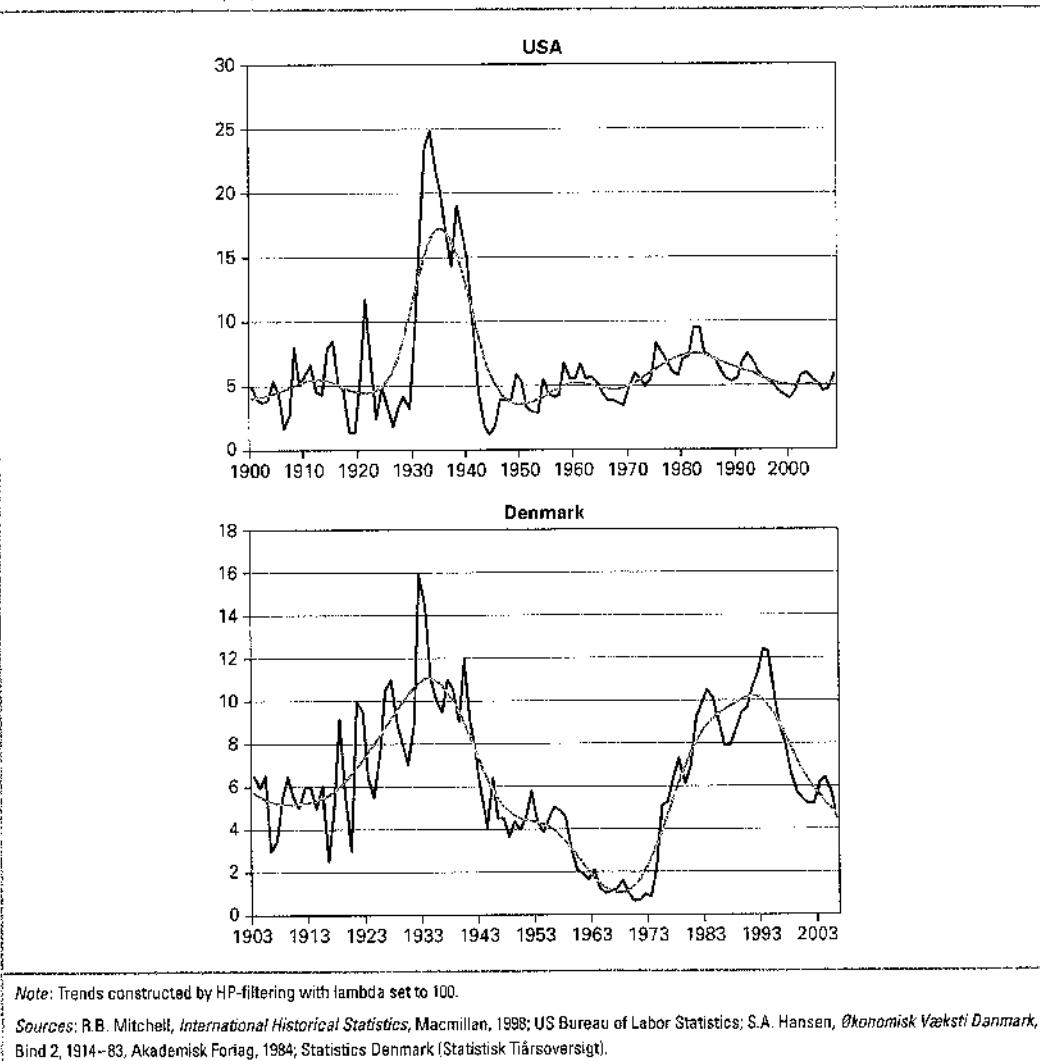
fall of 2008 implied negative values of GDP growth and increases in the rates of unemployment larger than the largest ones reported in the figure. Including the latest financial crisis period would only make the fluctuations appear larger.

The two main branches of macroeconomics are exactly divided by one being concerned with the trends of main economic time series, and the other with the considerable fluctuations around the trends:

THE SUBJECT MATTER OF MACROECONOMICS FOR THE LONG RUN AND FOR THE SHORT RUN



Macroeconomics for the long run is about understanding the **trends** in the most important macroeconomic series for GDP, consumption, investment, unemployment, etc., thus seeking explanations for long-run income levels and economic growth and so-called structural unemployment. Macroeconomics for the short run is about understanding the annual or quarterly **fluctuations** around trend in the most important macroeconomic series, thus seeking explanations for business cycles.

FIGURE 1.2 Annual unemployment rate in the USA, 1900–2008, and Denmark, 1903–2006

Note how each part of macroeconomic theory corresponds to each of the main questions asked in our definition of macroeconomics.

Most economists believe that an understanding of the trend requires a different type of explanation than an understanding of the fluctuations. The different macroeconomic models are formal expressions of these different explanations. The fundamental assumptions of the models for the long run and for the short run therefore differ, and hence the models themselves become different.⁴ But why should there be different macroeconomic theories for understanding the trends and for explaining the fluctuations? We will now consider this question in more detail.

4. Not all economists think that different assumptions are necessarily needed to explain short-run and long-run macroeconomic phenomena. Some believe that one model can be used for understanding both growth and fluctuations, since they do not see the fluctuations as coming (only) from exogenous shocks, but (also) from forces within the economic system itself. This hypothesis of *endogenous* business cycles is too advanced for the present course.

1.3 Macroeconomic theory for the short run

It may be illuminating to start with an example. Consider the Danish economy in 1993 and 1994 as illustrated in Fig. 1.3. In 1993 Danish real GDP was completely stagnant, but in 1994 it increased by about 5.5 per cent! This dramatic shift affected the rate of unemployment which dropped by almost two percentage points in one year.

Exogenous shocks

A standard macroeconomic explanation for this short-run fluctuation would start by saying that in 1994 the Danish economy was hit by a positive *exogenous shock*, that is, by a sudden event which is viewed as coming from outside the (Danish) economic system. Such an event could be either a *supply shock* like a sudden increase in the productivity of resources, or a *demand shock* like a sudden rise in domestic consumption, investment or exports rooted, for example, in more optimistic expectations concerning the future, or in a more expansionary fiscal or monetary policy, or in a sudden increase in the demand for Danish exports. Danish fiscal policy was in fact relaxed in 1993–94, and the real interest rate fell at that time. Moreover, the Danish export markets in Europe started to grow more quickly in 1994. Thus we may assume that the 1994 shock was mainly a positive demand shock.

To say that the sudden increase in Danish production was caused by an exogenous shock, basically an unexplained event, is not exactly deep. But note two things. First, economies are hit all the time by sudden events which are, perhaps, best considered as exogenous from the viewpoint of economic theory. For example, economists should not be concerned with explaining fluctuations in harvests due to shifting weather conditions, and probably they should not try to explain *all* sudden changes in the moods of consumers and investors. Focusing on the small economy of Denmark, a sudden increase in the demand for exports due to events in foreign countries should also be considered exogenous. Second, the occurrence of the shock is not the end of the story. A shock may be what *initiated* the change in economic activity, but it cannot itself explain all the subsequent economic reactions of households and firms. There is more explanation to do.

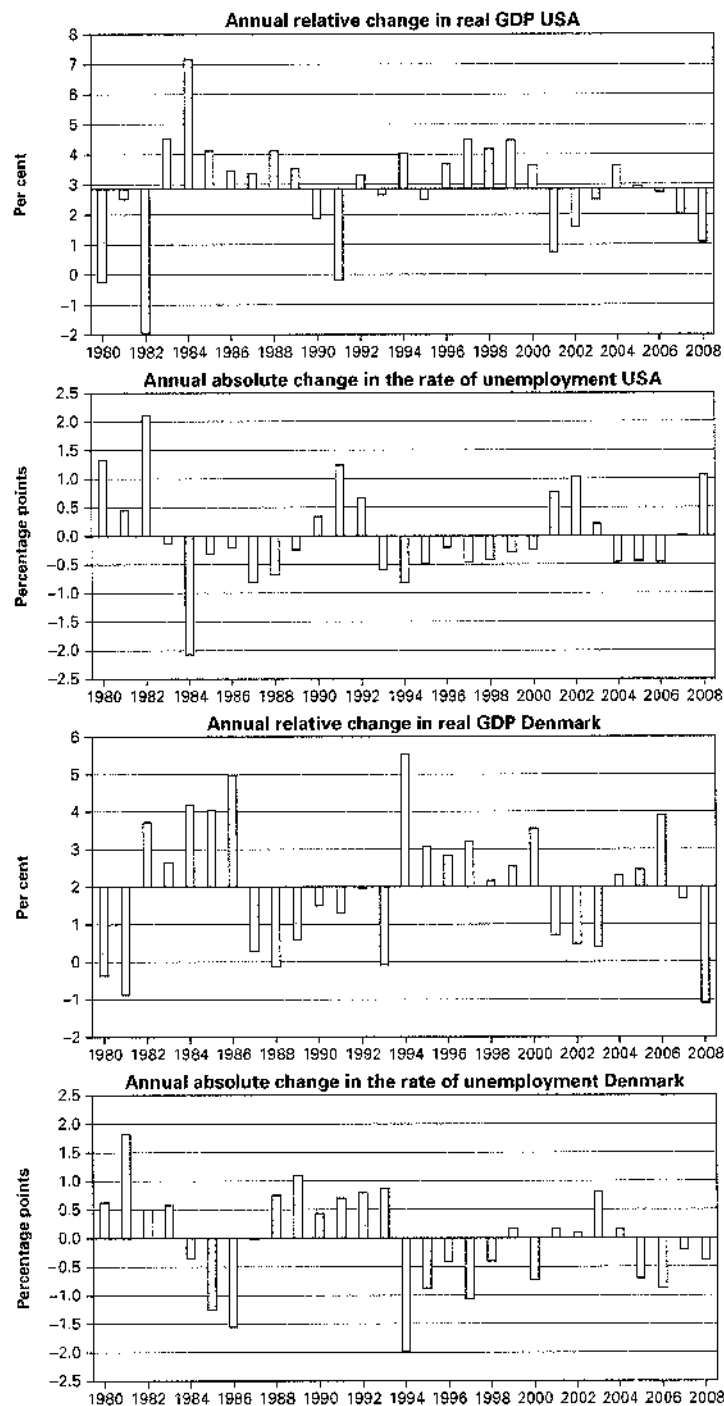
Nominal rigidities

For an increase in the aggregate demand for goods and services to lead to an increase in production, as observed in Denmark in 1994, it must have been profitable for Danish firms to increase their supply of output to accommodate the increase in demand. In the short run the capital stock is more or less given, so production can only increase if more labour is used. As more workers come to utilize the given capital stock, the marginal productivity of additional hours worked is likely to decline. If the marginal productivity of labour is indeed falling, it seems plausible that firms will only want to expand employment if the real wage falls. The real wage rate is defined as W/P , where W is the money wage rate and P is the price level. Presumably the higher demand for goods and services will lead to some increase in P . If the nominal wage rate W is rigid in the short run, this rise in prices will indeed cause a fall in the real wage, inducing firms to hire more labour to increase their supply of output.

Thus another key ingredient in our explanation of the fluctuation is the assumption of some *short-run nominal rigidity*, in this case a rigid money wage. The assumption that nominal wage rates are fixed for a certain period of time is quite realistic. Firms and workers do not renegotiate their wage rates every day or every month, because negotiation is a time-consuming process involving the risk of unpleasant and costly industrial conflict.

But experience also indicates that the nominal prices of most goods and services are only adjusted with certain time intervals. In an interesting empirical analysis of newsstand prices of American magazines, economist Stephen G. Cecchetti found that under an average general inflation of 4 per cent per year magazine prices were changed only every 6 years on average.

FIGURE 1.3 Annual relative change in GDP and absolute change in unemployment rate, USA and Denmark, 1980–2008



Note: The origins of the columns indicating relative changes in GDP are the average annual growth rates of GDP, which are 2.9 and 2.0 for the USA and Denmark, respectively. The actual annual growth rates are the distances from zero to the top/bottom.

Source: DECD, Economic Outlook Database.

This means that on average the real price of a magazine is eroded by about 25 per cent by inflation before the nominal price is changed.⁵

The rigidity of nominal magazine prices is not just a special case. In an empirical analysis of price rigidity, economist Alan Blinder asked a sample of business managers: 'How often do the prices of your most important products change in a typical year?' About 50 per cent of the managers responded that they changed their prices only once or less than once a year.⁶ Explaining why (most) firms do not immediately adjust their prices in response to changes in demand and cost is an intriguing issue to which we return in Section 1.5 below.

We argued above that a rigid nominal wage in association with a flexible, upward-adjusting nominal price could create the fall in real wages that would make it profitable for firms to supply more output in response to a positive demand shock. If a fall in the real wage is the typical reason why firms want to increase their output in reaction to a positive demand shock, we should expect to observe a negative relationship between output and real wages 'over the business cycle'. However, output often increases without a simultaneous decrease in real wages. It is therefore important to ask if an increase in the demand for output can induce firms to increase their supply even if *all* nominal prices are fixed in the short run (so the real wage does not fall). The answer is 'yes', provided that prices are *above* marginal costs before the demand shock hits the economy. In practice, most markets are characterized by *imperfect competition* where firms have some monopoly power enabling them to charge prices which are indeed above marginal costs. In that case they will be able to increase their total profit by increasing their output in response to an increase in demand, even if they have to keep their prices fixed temporarily.

The basic point is that short-run nominal price or wage rigidity can explain why an exogenous increase in *nominal* aggregate demand leads to a short-run increase in *real* output and employment. If nominal prices are fixed, all of the increase in demand will be reflected in a rise in real output, because imperfectly competitive firms will be happy to increase their supply as long as their (fixed) prices remain above marginal costs. And even if prices increase in response to higher demand, a rigid nominal wage means that the price hike will drive down the real wage which in turn will stimulate employment and output. In practice, both nominal wages and nominal prices are rigid in the short run, although to different degrees in different markets.

Expectational errors

So far our explanation for the short-run fluctuation in Danish output has relied on two ingredients: exogenous shocks and short-run nominal rigidities. But there is a third ingredient which is essential for a full understanding of the fluctuation: *expectational errors*. To see this, consider the Danish boom in 1994 and suppose that there was in fact some fall in the Danish real wage, as some prices increased in response to growing demand and nominal wages lagged behind. Faced with falling real wages, why were Danish workers nevertheless willing to increase their supply of labour, thereby enabling firms to expand employment and output? One possible answer is that trade unions in the heavily unionized Danish labour market had pushed real wages above the marginal disutility of work so that some workers were involuntarily unemployed prior to the demand boom. By definition, a worker who is involuntarily unemployed is willing to take a job even if the real wage falls below its current level (provided it does not fall too much).

But this hypothesis only begs the question why trade unions faced with growing labour demand would allow the real wage to fall. If unions were bargaining in 1993 to achieve a

5. Stephen G. Cecchetti: 'The Frequency of Price Adjustment: A Study of the Newsstand Prices of Magazines', *Journal of Econometrics*, 31, 1986, pp. 255–274.

6. Alan S. Blinder: 'On Sticky Prices: Academic Theories Meet the Real World', in N.G. Mankiw, ed., *Monetary Policy*, Chicago, University of Chicago Press, 1994.

certain real wage ω for 1994, why would they suddenly accept that the real wage fell below the target for 1994 at a time when labour demand was increasing? The most plausible answer is that the fall in real wages was *unintended* by unions. If unions had perfectly anticipated the positive 1994 demand shock and its effect on the price level, they would have bargained for a higher 1994 money wage rate to secure their target real wage ω , that is, they would have demanded a money wage satisfying $W = P \cdot \omega$.

However, since (most) wages had to be set *before* the occurrence of the shock, and since the shock was *not* perfectly foreseen, the negotiated money wage had to be based on the *expected* price level P^e which did not include the full inflationary effect of the shock: $W = P^e \cdot \omega < P \cdot \omega$. When the shock hit and prices increased above their expected level, unions were locked into their nominal wage contracts, and given the employer's right to hire more workers at the negotiated money wage, unions had to allow their members to supply additional labour even though the *realized* real wage, W/P , turned out to be lower than the *target* real wage, $W/P^e = \omega$.

This example illustrates our point that business fluctuations typically involve expectational errors, in this case errors made by workers (trade unions). In the case where some prices as well as money wages are rigid in the short run, an unanticipated shock will also cause some firms to err in their expectations. When firms pre-set their prices for a certain period, they will base their pricing decisions on their expected costs which will be influenced by the expected general price level P^e . When the unanticipated shock hits the economy, some firms (call them Group 1) will be just about to adjust their prices and will be able to account for the inflationary cost effect of the shock. But many other firms (Group 2) which have recently reset their prices will choose to maintain their existing prices for a while, even though the increase in the prices charged by Group 1 firms drives the costs of Group 2 firms above the previously expected level. As long as the shock does not push marginal costs above the pre-set prices, even Group 2 firms will want to expand their output to accommodate the unexpected rise in demand.

MODELLING ASSUMPTIONS OF MACROECONOMICS FOR THE SHORT RUN



Macroeconomic theory for the short run, intended to explain the economic fluctuations from year to year or from quarter to quarter, typically includes the following three modelling features:

1. exogenous shocks, i.e., sudden abrupt influences on the economy coming from changes in preferences, technology, or economic policies,
2. short-run nominal rigidity, i.e., some period after the occurrence of a shock during which some prices and/or wages are sticky, and
3. expectational errors, i.e., a period after the occurrence of a shock during which some prices are different from what was expected before the shock.⁷

7. Some short-run macroeconomic theories do not assume price rigidity, e.g. the theory of Milton Friedman presented in 'The Role of Monetary Policy', *American Economic Review*, 58, 1968, pp. 1–17, and related more formal contributions such as Robert E. Lucas, Jr: 'Expectations and the Neutrality of Money', *Journal of Economic Theory*, 4, 1972, pp. 103–124, or the real business cycle theory explained in Section 18.4. A main point in these theories is that fluctuations can be understood within a framework that only assumes shocks and possibly expectational errors. Hence one cannot conclude from the mere observation of fluctuations that wages or prices are rigid: fluctuations are theoretically compatible with all wages and prices being fully adjusted all the time, as we shall see in Chapters 17 and 18. In fact, a main point of real business cycle theory is that one and the same class of economic models can be used for understanding both long-run growth issues and short-run issues of fluctuations. The models in this class are much like the growth models to be dealt with in subsequent chapters. The only relevant element to add to the growth models to make them also models of fluctuations is, according to this view, exogenous supply shocks. We find this theoretically interesting, but today it is widely accepted that short-run price and wage rigidities are indeed important for understanding the economy's short-run reactions to shocks and that demand shocks are important.

1.4 Macroeconomic theory for the long run

While exogenous shocks, temporarily sticky wages or prices and erroneous expectations are required for an understanding of the changes from year to year in unemployment and GDP, most economists think that these features are best disregarded when we try to explain the 'gravity level' of the rate of unemployment and the trend-wise gradual growth of GDP over long periods. The smooth trend curves in Figs 1.1 and 1.2 could not possibly reflect a succession of random shocks over the more than 100 years considered. By definition, shocks have to go in opposite directions from time to time. If technology improves constantly each year to imply a 2 per cent increase in GDP per head, then this annual shift in technology should not be considered as a shock, but rather as a foreseeable gradual movement. Moreover, although nominal wages and prices may be sticky in the short run, they do adjust in the longer run. In macroeconomics for the long run we therefore abstract from the three features which define short-run macroeconomics.

Long-run modelling: the basic assumptions

In other words, *macroeconomic theory for the long run*, intended to explain the trend-wise movements of main economic variables around which the year-to-year fluctuations occur, portrays the economy *as if exogenous shocks do not occur*, i.e., the economic fundamentals like preferences and technology develop in a smooth and foreseeable way over time; *prices are fully adjusted* in all periods in accordance with the economy's full long-run price flexibility; and *expectations are correct* all the time.

You should carefully note the 'as if' in this definition. If the assumptions of macroeconomics for the short run give the right picture, then in (almost) every year the economy will be reacting to shocks, with prices still not fully adjusted and expectational errors still prevailing. This is because new shocks occur all the time. Nevertheless, certain phenomena may be better understood by considering the economy *as if* shocks did not occur, prices were always fully adjusted, and expectations always correct. Among such phenomena we include the long-run growth performance and the long-run gravity level of unemployment. Thus macroeconomic models for the long run describe the underlying long-run equilibrium towards which the economy is gravitating, even though recurring shocks and the time-consuming adjustment to these shocks imply that the economy is never exactly in this long-run equilibrium.

To put it another way, in explaining how economic growth in Denmark could change from zero per cent in 1993 to 5.5 per cent in 1994, and unemployment at the same time be significantly reduced, we have to allow for exogenous shocks, price rigidities and expectational errors. In an explanation of the average annual GDP growth of 2.9 per cent over the period from 1950 to 2004, or of the average rate of unemployment of 5.7 per cent over the same period, it is better to abstract from these three complicating features to focus attention on the fundamental forces governing the long-run evolution of the economy.

MODELLING ASSUMPTIONS OF MACROECONOMICS FOR THE LONG RUN



In macroeconomic theory for the long run, intended to explain the trends of economic variables over the long run, the economy is analysed *as if*:

1. the economic fundamentals evolve smoothly and predictably over time (no exogenous shocks),
2. prices are adjusted in accordance with the economy's full long-run price flexibility (no short-run nominal rigidity), and
3. expectations are correct (no expectational errors).

Static versus dynamic models

A macroeconomic model for the long run can be a single-period *static* model. This may seem surprising, but sometimes it is useful to focus on a single period which is viewed as an 'end period' in the sense that no new shocks have occurred for a long time and the economy has finished all its adjustments. One can then describe the state of the economy in the end period. The purpose is to characterize the equilibrium towards which the economy is tending in the long run, without complicating the theory with an explicit analysis of the dynamic process which takes the economy to the long-run equilibrium. The models of long-run structural unemployment presented in Part 4 of this book will be of this nature.

By contrast, *dynamic* models for the long run describe the process of capital accumulation explicitly, and typically they also describe the evolution of other important stock variables such as the labour force, the stock of natural resources, etc. These models, also called growth models, always contain the following dynamic link between the current flow of investment and the increase in the stock of capital:

$$K_{t+1} - K_t = I_t - \delta K_t, \quad (1)$$

where K_t is the amount of capital available in period t , I_t is gross investment in period t , and δ is the rate at which capital depreciates, $0 < \delta < 1$. In a closed economy the level of gross investment I_t must equal gross savings S_t . For a closed economy the capital accumulation equation therefore becomes:

$$K_{t+1} - K_t = S_t - \delta K_t. \quad (2)$$

Thus savings play a central role in growth models. In many models it is simply assumed that households always save an exogenous fraction s of total income so that in any period t we have $S_t = sY_t$. The growth models using this assumption are called Solow models.⁸

In more advanced growth models savings are derived from maximization of utility functions under appropriate budget constraints. We leave these more complicated models for a later macroeconomics course. Solow models are well suited and widely used for understanding many growth issues. However, for analysing the effects of economic policies, these models have the shortcoming that they do not contain utility functions and therefore do not allow an explicit evaluation of the welfare consequences of alternative economic policies. In Solow models one will just have to look at the policy implications for economic variables like GDP or the level of consumption, and simply assume that it is 'good' to have high per capita income or consumption.

1.5 Real rigidities and the natural rate in macroeconomics for the long run

The assumption in long-run macroeconomics that all wages and prices are fully adjusted in all periods means that there are no *nominal* rigidities in the long term. But there may well be permanent *real* rigidities preventing real prices and wages from adjusting to the values which would prevail under perfect competition. The economy's long-run wage and price flexibility can be different from the flexibility assumed in the traditional model of perfect competition. We will now go through a simplified example to illustrate what we mean. For concreteness, we will show how real rigidities may arise from the market power of trade unions, but we emphasize that even if unions are weak or non-existent, there are

8. They have this name after the Nobel Prize Laureate Robert M. Solow who made fundamental contributions to growth theory. Some of Solow's contributions are the subjects of Chapters 3 and 5 below.

other mechanisms which may cause significant real rigidities, as our analysis in Chapter 11 will make clear.

Consider an economy divided into many different sectors, each represented by a firm producing a differentiated product, and each having an industry trade union which controls the supply of labour to firms in the sector. Suppose for simplicity that labour is the only variable factor of production and that it takes one unit of labour to produce one unit of output in each sector. The average and marginal cost of production for the individual firm will then be equal to the wage rate for its industry. We assume that the representative firm of each sector faces a demand curve with a constant price elasticity of demand and that this elasticity is the same in all sectors. It is well known from microeconomics that when the price elasticity of demand is constant, a profit-maximizing firm will set its price as a constant mark-up over its marginal cost. Hence, if the average nominal wage level across sectors is W , the average price level P may be specified as:

$$P = m^P W, \quad m^P > 1, \quad (3)$$

where m^P is the mark-up factor of the representative firm.

Now consider the trade union monopolizing the supply of labour to a particular sector i . To simplify, suppose the union is so strong that it can effectively dictate the nominal wage rate W_i to be paid by the firm representing sector i . The trade union sets the wage rate with the purpose of maximizing the objective function Ω where:

$$\Omega = \left(\frac{W_i}{P} - v \right) L_i. \quad (4)$$

Here L_i is the level of employment in sector i , W_i/P is the real wage for union members employed in firm i , and v is the real income which members are able to earn elsewhere if they fail to find a job in sector i . The variable v is sometimes referred to as the workers' 'outside option' and will be specified below. Since $W_i/P - v$ is the rent (surplus) which the individual union member earns by being employed in firm i rather than having to look for a job elsewhere, the magnitude Ω is the total rent that union members gain from employment in firm i . In Chapter 12 we shall explain in detail why it is reasonable to assume that the trade union wants to maximize this total rent; for the moment we ask you to accept this plausible assumption.

When setting the wage rate W_i , the union must take into account that a higher wage will induce the firm to hire fewer workers. If the union charges an average wage rate so that $W_i/W = 1$, the firm in sector i will have the same cost level as its competitors in other sectors and will experience an average volume of sales. The union may then expect to gain a proportional share of total employment, so employment in sector i will be L/n , where L is aggregate employment and n is the number of different sectors. But if the union for sector i raises its relative wage rate, W_i/W , its employer will have to raise his relative output price, leading to a lower volume of sales and employment in the sector. Noting from (3) that $W = P/m^P$, we may thus specify the demand for labour in sector i as:

$$L_i = \frac{L}{n} \left(\frac{W_i}{W} \right)^{-\sigma} = \frac{L}{n} \left(\frac{m^P W_i}{P} \right)^{-\sigma}, \quad \sigma > 1, \quad (5)$$

where the parameter σ measures the numerical wage elasticity of labour demand. Since the trade union for sector i is small relative to the rest of the economy, it takes the outside option v and the general price level P as given. By setting the nominal wage rate, W_i , the union thus implicitly chooses its real wage, W_i/P . Using (5) to eliminate L_i from (4), and differentiating the resulting expression for Ω with respect to W_i , we obtain a necessary (first order) condition for the optimality of the wage rate set by the union. Since the log-function is increasing we may as well differentiate the log of Ω , which is convenient here:

$$\frac{d \ln \Omega}{d W_i} = \frac{1}{\frac{W_i}{P} - v} \frac{1}{P} - \sigma \frac{1}{W_i} = 0 \Leftrightarrow$$

$$\frac{W_i}{P} = m^w v, \quad m^w \equiv \frac{\sigma}{\sigma - 1} > 1. \quad (6)$$

We see that the union will set the real wage as a constant mark-up over the outside option. Let us take a closer look at this option. If the unemployment rate is u , the representative union member has a probability $1 - u$ of finding another job if he has to look for work outside sector i . In that case he may expect to earn the average real wage W/P . There is also a probability u that the union member will remain unemployed if he does not obtain a job in sector i . In that situation the worker will receive a real unemployment benefit, b , assumed to be lower than the average real wage. The outside option, defined as the real income a worker may expect to earn outside sector i , may thus be specified as:

$$v = (1 - u) \frac{W}{P} + ub. \quad (7)$$

From (6) and (7) we may derive a simple theory of long-run unemployment which will help to clarify the concept of real rigidities. Inserting (7) into (6), and denoting the real wage rates inside and outside firm i by w_i and w , respectively, we get:

$$w_i = m^w [(1 - u)w + ub], \quad w_i \equiv \frac{W_i}{P}, \quad w \equiv \frac{W}{P}. \quad (8)$$

Assuming that the union mark-up factor m^w is the same across sectors, it follows from (8) that trade unions will charge the same wage rate in all sectors. Hence we may set $w_i = w$ in (6) and solve for w to get:

$$w = \frac{m^w u}{1 - m^w (1 - u)} b = \frac{1}{1 - \frac{m^w - 1}{m^w} \frac{1}{u}} b, \quad (9)$$

where we assume $u > (m^w - 1)/m^w$ for a meaningful expression. Equation (9) is the economy's so-called *wage curve* showing how the real wage claimed by unions depends on the unemployment rate and the rate of unemployment benefit: under the condition just stated, the real wage claim falls if unemployment goes up.

In a macroeconomic equilibrium the real wage claimed by unions must be consistent with the real wage implicitly offered by firms through their price setting. According to (3), firms implicitly offer the real wage $w \equiv W/P = 1/m^p$. Inserting this into (9) and solving for u , we get the *long-run equilibrium unemployment rate*, also called the natural (or the structural) rate of unemployment:

$$\bar{u} = \frac{\frac{m^w - 1}{m^w}}{1 - m^p b} = \frac{m^w - 1}{m^w (1 - m^p b)}. \quad (10)$$

It follows immediately that if $m^w = 1$, then $\bar{u} = 0$. In other words, if unions have no market power and labour markets are characterized by perfect competition (a situation arising as the demand elasticity σ goes to infinity), the natural rate of unemployment is zero. You may check for yourself that the equilibrium unemployment rate will be positive and smaller than one provided that $m^w m^p b < 1$ or, equivalently, $b/w < 1/m^w$. Hence, if unions do have market power, $m^w > 1$, and the 'replacement ratio' b/w offered by the system of unemployment insurance is not 'too' high, we get a strictly positive and meaningful solution for the equilibrium

unemployment rate. (Our assumption $m^w m^p b < 1$ implies $w > b$ as well as $\bar{u} > (m^w - 1)/m^w$ as assumed above, which you should check.) The monopoly power of the trade unions implies a real rigidity in the sense that an excess supply of labour (i.e., unemployment) can exist without creating any downward pressure on real wages.

Note that a positive unemployment rate emerges even though we did not assume any *nominal* rigidities: the nominal variables W and P can be freely scaled up or down in the equations above.

The unemployment problem is rooted in the market distortions reflected in the mark-up factors m^p and m^w . If there were no unions and labour markets were perfectly competitive, workers could not drive wages above their outside option. According to our simplified model equilibrium unemployment would then be zero, as you can see by setting $m^w = 1$ in (10). But unions *do* exist and *do* have market power.⁹ In the situation where $m^w > 1$, the unemployment generated by union monopoly power will be *exacerbated* by imperfect competition in product markets. This is because weaker product market competition implies a higher value of the price mark-up, m^p , and according to (10), this will increase the value of $\bar{u}$: when firms drive down the level of real wages by pushing prices above marginal costs, the amount of involuntary unemployment must rise to bring union wage claims down in line with the real wages implicitly offered by firms through their monopolistic price setting. We also see from (10) that a higher level of unemployment benefits will raise the equilibrium unemployment rate when unions have market power.

When economists speak of 'real rigidities', they often mean that market imperfections permanently distort the real prices and real wages claimed by firms and workers away from the competitive relative (real) prices which would ensure full resource utilization. Even in the long run, when all nominal wages and prices have had time to fully adjust to their desired levels, these structural market distortions will persist, leaving the economy in a state of permanent unemployment. Our analysis suggests that the degree of real rigidity in the economy may be measured by the size of the parameters m^w , m^p and b which contribute to equilibrium unemployment. Alternatively, we might simply measure the degree of real rigidity by the size of the equilibrium unemployment rate, since we have seen that $\bar{u} = 0$ in the benchmark case of perfect competition.

According to established tradition in macroeconomics, the long-run equilibrium unemployment rate implied by the economy's *real* rigidities is called the *natural rate*. This is the rate of unemployment emerging when all relative prices have fully adjusted in accordance with the economy's long-run wage and price flexibility. In line with this tradition, we will use the term 'natural rate of resource utilization' to denote the rate at which factors of production are utilized in long-run equilibrium when the economy has exhausted its potential for price adjustment. By definition, there are no real rigidities if perfect competition prevails, since real (relative) prices in a competitive economy adjust until the demand for each resource equals the full supply of that resource. This just means that the natural rate of resource utilization is 100 per cent. If real rigidities do exist, the natural rate of resource utilization will be less than 100 per cent.

We have suggested that the degree of real rigidity may be measured by the *level* of the natural unemployment rate. We should add that economists sometimes find it fruitful to work with a slightly different concept of real rigidity. In this alternative definition, the degree of real rigidity is measured by the *responsiveness* of real wages and real prices to a short-run *change* in the unemployment rate away from the natural rate. If this responsiveness is low, the degree of real rigidity is said to be high. In Section 1.6 we will adopt this alternative notion of real rigidity which will turn out to be useful for explaining short-run nominal rigidities. At any rate, real rigidity in a broad sense refers to the fact that the economy's degree of relative price flexibility is less than it would be in an ideal world of perfect competition.

9. You may ask why rational workers would want to form trade unions if unions create involuntary unemployment. Chapter 12 will offer an answer to this.

The crucial role of the supply side in long-run modelling

As indicated by the simple example in Eq. (10), the natural rate of employment, $1 - \bar{u}$, is given from the *supply side* of the economy, since the parameters m^w , m^p and b are characteristics of the economy's supply side structures. This has an important implication: in macroeconomic models for the long run, where wages and prices are assumed to be fully adjusted in all periods, output is determined solely from the supply side. In any given period there is a certain labour force L , and a certain (predetermined) capital stock K . Output is then completely constrained from the supply side, since it cannot exceed the volume which can be produced by means of 'natural' labour input $(1 - \bar{u})L$ and the predetermined capital input K (here assuming that the natural rate of utilization for capital is one).

By contrast, in short-run macroeconomic models nominal wage and price rigidities and/or expectational errors may cause employment to deviate from its natural rate. Hence we do not have the simple supply side determination of employment just described. Instead, employment is also influenced by the aggregate *demand* for goods and services. Thus, in long-run macroeconomic models employment always corresponds to the natural rate, whereas in short-run models employment is determined also from the demand side and fluctuates around the natural rate.

THE NATURAL RATE AND OUTPUT DETERMINATION IN THE LONG RUN AND IN THE SHORT RUN



In macroeconomics for the long run, long-run real price rigidities may prevail and imply that natural rates of resource utilization are smaller than one, so there can be structural unemployment in the long run. Output is determined from the supply side as what can be produced from available resources when these are utilized at their natural rates.

In macroeconomics for the short run, short-run nominal price rigidities can prevent price adjustment from continuously ensuring that resources are utilized at natural rates. Therefore output is not determined from the supply side alone, but rather by an interaction between supply and demand. In the short run output can therefore fluctuate around the natural rate of output.

1.6 Explaining the nominal rigidities of macroeconomics for the short run

We noted earlier that some form of nominal rigidity typically plays an important part in the explanation of short-run business fluctuations. As we shall demonstrate in this section, the phenomenon of *real* rigidity discussed in the previous section may help us understand how short-run *nominal* rigidity may be compatible with rational economic behaviour. Specifically, we will show that when target real wages are not very responsive to changes in the level of economic activity, even very small costs of nominal wage or price adjustment may be sufficient to induce rational agents to abstain from adjusting nominal wages or prices in the short run in response to fluctuations in economic activity.

It is not unreasonable to assume that there are some small costs of adjusting nominal wage rates. For example, union leaders may have to spend some time sitting down at the bargaining table with the employer to write a new wage contract. Moreover, when the demand for labour changes, wage setters may have to spend some resources estimating the exact magnitude of the change if they want to be sure that they set the optimal wage rate. Similarly, firms may incur small costs in resetting their nominal prices. For instance, they may have to print new catalogues or spend other resources communicating the new prices to the market.

The costs of adjusting nominal prices or wages are often referred to as 'menu costs', like the costs to a restaurant of having to print a new menu card. In most cases such costs are likely to be quite small, and hence they are usually neglected in economic models. But if we

can show that even very small menu costs may be sufficient to induce economic agents not to adjust nominal wages or prices in response to even considerable shocks, we have a potential explanation for the nominal rigidities which seem to exist in the real world.¹⁰

Explaining nominal wage stickiness

We start by considering the circumstances in which nominal wage rigidity may be the outcome of optimizing economic behaviour. Our strategy is to measure the loss incurred by individual wage setters if they do *not* adjust their nominal wage rate in reaction to a change in the unemployment rate of a realistic magnitude. If this loss is very small, then the privately optimal behaviour may be to leave the wage rate unchanged to avoid the cost of wage adjustment.

We base our analysis on the simple model of wage formation developed in the previous section, adding to this a shock and small menu costs. For convenience, we now assume that the system of unemployment insurance offers a fixed compensation relative to the average wage level, $b \equiv cw$, where the replacement rate $c < 1$ is a constant. Recalling that $w \equiv W/P$, the outside option (5) may then be written as a function of the unemployment rate:

$$v = v(u) = [1 - (1 - c)u]w. \quad (11)$$

A higher unemployment rate implies a lower value of the outside option. We do not state explicitly that the outside option v is also a function of the average economy-wide real wage w , since we know from (3) that w will remain constant at the level $1/m^p$ as long as nominal prices are not rigid. Inserting (11) into (6) and remembering that $w_i \equiv W_i/P$, we get the optimal real wage which the union would choose in the *absence* of any costs of wage adjustment:

$$w_i = m^w v(u) = m^w [1 - (1 - c)u]w. \quad (12)$$

Without loss of generality, we may choose our units of measurement such that the total labour force is normalized to unity, implying $L = 1 - u$. Hence, from Eq. (5), labour demand in each sector will be: $L_i = [(1 - u)/n](m^p w_i)^{-\sigma}$. Substituting this expression into our expression for Ω in (4), and also inserting the expression (11) for v , we may write the objective function for the union organizing the workers in sector i as:

$$\Omega(w_i, u) = (w_i - v(u)) \overbrace{\left(\frac{1-u}{n} \right)}^{L_i} (m^p w_i)^{-\sigma}. \quad (13)$$

You may want to check again that the first-order condition for maximization of Ω with respect to w_i gives (12). We can now analyse whether the union in a specific sector would actually want to adjust its own real wage w_i relative to the average real wage if the overall unemployment rate were to change due to a shock. If the union wanted to adjust its real wage it would do so by changing its *nominal* wage rate W_i , since this is the only way it can affect its real wage rate. And if all unions had an incentive to adjust their own real wage relative to the market real wage, this would initiate an economy-wide adjustment of nominal wage rates, so nominal wage rigidity would be absent, even though real wage rates would stay constant due to the mark-up price setting of firms.

10. A seminal contribution to the menu cost theory of nominal rigidity was made in the highly readable article by N. Gregory Mankiw: 'Small Menu Costs and Large Business Cycles: A Macroeconomic Model of Monopoly', *Quarterly Journal of Economics*, 100, 1985, pp. 529–539. Later on, the menu cost theory was elaborated by Mankiw and several other so-called New Keynesians, including Laurence Ball and David Romer: 'Real Rigidities and the Non-Neutrality of Money', *Review of Economic Studies*, 57, 1990, pp. 183–203.

Suppose that the union in sector i has initially set its wage optimally given some initial level of unemployment $\bar{u}$, so that the real wage rate is $\bar{w}_i = m^v v(\bar{u})$. Suppose further that an exogenous shock drives unemployment to a new and higher level u . In the absence of any costs of nominal wage adjustment, the union would then want to adjust its nominal wage rate to secure a real wage of $w_i = m^v v(u)$, as given by (12). If it chooses *not* to change its wage rate, leaving it at the initial level $\bar{w}_i$ to avoid the costs of wage adjustment, the union will suffer the following utility loss, UL , excluding adjustment costs:

$$UL = \Omega(w_i, u) - \Omega(\bar{w}_i, u) \quad (14)$$

Since the union's objective function (and hence UL) is specified in terms of real income, the utility loss in (14) is directly comparable to the fixed real cost of nominal wage adjustment, denoted by C . Hence the union will want to maintain the real wage rate $\bar{w}_i$ if $UL < C$, and in that case its nominal wage will be rigid. To estimate how high C will have to be to prevent wage adjustment, it is convenient to take a second-order Taylor approximation of $\Omega(\bar{w}_i, u)$, calculated around the new optimum wage rate w_i :

$$\Omega(\bar{w}_i, u) \approx \Omega(w_i, u) + \overbrace{\frac{\partial \Omega(w_i, u)}{\partial w_i}}^{\approx 0 \text{ in optimum}} (\bar{w}_i - w_i) + \frac{1}{2} \frac{\partial^2 \Omega(w_i, u)}{(\partial w_i)^2} (\bar{w}_i - w_i)^2. \quad (15)$$

Note that since w_i is the new *optimal* wage rate (in the absence of adjustment costs), it must satisfy the first-order condition, $\partial \Omega / \partial w_i = 0$. Hence we may approximate the utility loss in (14) by:

$$UL \approx -\frac{1}{2} \frac{\partial^2 \Omega(w_i, u)}{(\partial w_i)^2} (\bar{w}_i - w_i)^2. \quad (16)$$

Essentially (16) says that the trade union incurs no *first-order* utility loss of not adjusting its wage rate, but only a second-order loss. The second derivative in (16) may be calculated from (13). Doing this, using the definition $m^v \equiv \sigma/(\sigma - 1)$ plus the fact that (6) implies $(w_i - v)/w_i = 1/\sigma$, and expressing UL as a fraction of the union's total wage bill $w_i L_i = w_i((1 - u)/n)(m^v w_i)^{-\sigma}$, we find that:¹¹

$$\frac{UL}{w_i L_i} \approx \left(\frac{\sigma - 1}{2} \right) \left(\frac{\bar{w}_i - w_i}{w_i} \right)^2. \quad (17)$$

11. From (13) we have

$$\begin{aligned} \frac{\partial \Omega}{\partial w_i} &= \left(\frac{1 - u}{n} \right) [(m^v w_i)^{-\sigma} - \sigma m^v (w_i - v)(m^v w_i)^{-\sigma-1}] \Rightarrow \\ \frac{\partial^2 \Omega}{(\partial w_i)^2} &= \sigma m^v \left(\frac{1 - u}{n} \right) [m^v (\sigma + 1)(w_i - v)(m^v w_i)^{-\sigma-2} - 2(m^v w_i)^{-\sigma-1}] \\ &= \sigma m^v \left(\frac{1 - u}{n} \right) (m^v w_i)^{-\sigma-1} \left[(\sigma + 1) \left(\frac{w_i - v}{w_i} \right) - 2 \right]. \end{aligned}$$

Inserting $m^v = \sigma/(\sigma - 1)$ and $(w_i - v)/w_i = 1/\sigma$ into the latter expression, we get:

$$\frac{\partial^2 \Omega}{(\partial w_i)^2} = -\sigma \left(\frac{m^v}{m^v} \right) \left(\frac{1 - u}{n} \right) (m^v w_i)^{-\sigma-1}.$$

Substituting this expression into (16), dividing by $w_i L_i = w_i((1 - u)/n)(m^v w_i)^{-\sigma}$, and using the definition of m^v again, we end up with (17).

Before the shock to unemployment, the union had set its wage rate $\bar{w}_i$ optimally in accordance with (12) given $\bar{u}$, and w_i is also given by (12) at u . Since the general wage level is the same in both situations ($w = W/P = 1/m^p$), it follows that:

$$\frac{\bar{w}_i - w_i}{w_i} = \frac{(1-c)(u - \bar{u})}{1 - (1-c)u}. \quad (18)$$

Not surprisingly, we see from (17) and (18) that the larger the rise in unemployment, $u - \bar{u}$, and hence the larger the fall in the value of the outside option of union members, the larger is the union's utility loss from not adjusting the wage rate.

Armed with Eqs (17) and (18), we can calculate how high the costs of nominal wage adjustment will need to be to prevent such adjustment, assuming a set of parameter values which generate a realistic initial unemployment rate. First we note that when $b = cw = c/m^p$, the expression for the natural unemployment rate given in (10) simplifies to

$$\bar{u} = \frac{m^w - 1}{m^w(1-c)} = \frac{1}{\sigma(1-c)}. \quad (19)$$

As an example, suppose that the replacement rate c is equal to 0.5 (not unrealistic for the average worker in many OECD countries) and that the economy is initially in long-run equilibrium at an unemployment rate equal to 5 per cent. Inserting $\bar{u} = \bar{u} = 0.05$ and $c = 0.5$ into (19), it follows that $\sigma = 40$. Finally, suppose that the increase in the unemployment rate is two percentage points, $u - \bar{u} = 0.02$, implying a large 40 per cent increase in unemployment. Inserting these parameter values into (17) and (18), one finds that $UL/w_i L_i \approx 0.002$ or 0.2 per cent. In other words, the costs of wage adjustment – broadly interpreted to include the costs and inconveniences of rewriting wage contracts plus the resource costs of re-estimating the optimal wage – only have to amount to about one-fifth of 1 per cent of the wage bill to make it suboptimal for the individual union to reduce its nominal wage claim, despite the sharp rise in unemployment. If the rise in the unemployment rate is only 1 percentage point (still a substantial 20 per cent increase in unemployment), our parameter values imply that the 'menu costs' of wage adjustment only have to make up a tiny 0.05 per cent of the wage bill to generate nominal wage rigidity. Adjustment costs of such small magnitude certainly do not seem unlikely.

To sum up, even very small menu costs may cause the individual trade union's nominal wage claim to be rigid in the short run, and as a consequence the general nominal wage level will be rigid.

The link between real and nominal rigidity

Nominal and real rigidities are linked in the sense that a stronger degree of real rigidity can imply more nominal rigidity: the term $(\bar{w}_i - w_i)/w_i$ appearing in (17) measures how much the union would like to change its *real* wage rate if adjustment were costless, given a change in overall unemployment from $\bar{u}$ to u . Hence $(\bar{w}_i - w_i)/w_i$ may be seen as an inverse measure of the degree of real wage rigidity. The smaller this magnitude, the weaker is the desire for real wage adjustment in response to fluctuations in unemployment, so the higher is the degree of real wage rigidity, and vice versa. Eq. (18) shows that in this particular setting the degree of real rigidity is determined by the replacement ratio, and the larger c is, the stronger is the real rigidity. And the greater the degree of real wage rigidity, the smaller are the menu costs needed to prevent *nominal* wage adjustment, so the higher is the degree of nominal wage rigidity. On the other hand, if $(\bar{w}_i - w_i)/w_i$ is large, reflecting a low degree of real rigidity, small menu costs are unlikely to prevent nominal wage adjustment, according to (17). In short, a relatively high degree of real wage rigidity tends to create relatively strong nominal wage rigidity.

Explaining nominal price rigidity

The menu cost theory may also explain the rigidity of nominal *prices*. Obviously, if firms set their prices in accordance with our mark-up price setting equation $P = m^p W$, and if the nominal wage rate is rigid, it follows that nominal prices will also be rigid.

But even if there is *no* nominal wage rigidity, there may still be nominal price rigidity. If we assume zero costs of nominal wage adjustment so that nominal wages are fully flexible, the labour market model set up above still implies considerable *real* wage rigidity for reasonable parameter values. This means that wage setters (unions) will not want to change nominal and real wage rates very much when economic activity changes, despite the fact that wage adjustment is costless. Consequently firms will experience only moderate fluctuations in their real wage costs over the business cycle. One can construct plausible examples in which this rigidity in the real production costs of firms will generate nominal price rigidity even when the menu costs of price changes are very small.

This parallel between the lack of incentives for wage and price adjustment may be explained as follows: in our model of union wage setting the outside option is the union's real opportunity cost of 'selling' an extra job to the owner of firm i through wage moderation (since the outside option is the real income a union member could have earned elsewhere in the labour market). Because the outside option is rather insensitive to changes in unemployment, the union loses little by not adjusting its wage rate. Similarly, if the firm's real labour cost of producing and selling an extra unit of output is insensitive to the business cycle, it has little incentive to adjust its real price, and hence it will often abstain from adjusting its nominal price in order to save menu costs.

Since the individual firm takes the prices set by all other firms as given, a decision by firm i to change its nominal price P_i is also a decision to change its real (relative) price, P_i/P . Hence nominal price rigidity is a reflection of real price rigidity, just as nominal wage rigidity arises from real wage rigidity.

Individual versus social costs of nominal rigidities

It should be emphasized that although the cost to the individual union (the private cost) of not adjusting its wage rate is likely to be small, the *social* cost of nominal wage rigidity may be substantial. The reason why the individual union estimates such a small gain from adjusting its own nominal wage is that it takes all other wages and prices as given. But if all unions could be induced to cut their nominal wage rates simultaneously in response to a drop in aggregate employment, they would collectively drive down the rate of inflation (via the price-setting equation $P = m^p W$), and as we shall see in Book Two, this will tend to increase real aggregate demand. In this way it might be possible to limit the fall in output and employment significantly. Take our first numerical example above: if a coordinated downward adjustment of wages and prices could secure an expansion of aggregate demand which would cut the two percentage point rise in unemployment in half, total output and real income would rise by more than one percentage point relative to the baseline scenario where no nominal adjustment takes place, given the assumption underlying our price setting equation that the production function is linear in labour input.¹² Against this significant income gain must be set the small menu cost of wage adjustment which is only 0.2 per cent of the wage bill, and hence an even smaller percentage of total GDP. Thus, since nominal wage rigidity causes a substantial loss of output, its social cost is many times as large as the perceived private cost of non-adjustment of nominal wages. For exactly the same reasons, the social cost of nominal price rigidity is likely to be many times as great as the private cost perceived by individual price setters. This is why the problem of nominal rigidity should be taken seriously.

12. With a linear production function, total GDP (Y) after the shock to unemployment is $Y = 1 - u = 1 - 0.07 = 0.93$, so if unemployment were to rise by one percentage point less, output would rise by $dY/Y = du/(1 - u) = 0.01/0.93 = 0.011$ relative to the baseline scenario where unemployment rises by two percentage points.

THE SMALL MENU COST FOUNDATION OF NOMINAL WAGE AND PRICE RIGIDITY



A menu cost associated with changing nominal prices (and wages) can, even if it is small, prevent price-setting imperfect competitors from adjusting their nominal prices in response to a shock even if this shock is of considerable size. The reason is that if the price is set optimally at the outset, a price adjustment will only have a second- (and higher-) order influence on the objective of the price setter, and when there are real rigidities these second- and higher-order effects will be small.

Although the cost (excluding menu costs) of not adjusting the nominal price in case of a shock is small for each individual price setter, given that other price setters also keep their nominal prices unchanged, the cost of nominal rigidities where many price setters are subject to small menu costs can be large, since there could be much to gain for everybody from a coordinated price reaction to the shock.

We should stress that the menu cost theory outlined here is only intended to explain *short-run* nominal rigidities. Over time as shocks accumulate, agents who keep their wages or prices constant are likely to be pushed ever further away from their optimal wages and prices. Eventually they will therefore find it optimal to pay the menu cost to realign their relative price position: in the long run wages and prices *do* adjust.

The role of shocks, expectational errors and nominal rigidities: a simple restatement

Our analysis of nominal rigidities assumed that unemployment could somehow deviate from its natural rate, but we did not explain how. We will now do so, since this will give us an opportunity to highlight the role of expectational errors. Going back to equations (4) and (6), we see that these equations assume that wage setters correctly anticipate the general level of wages and prices when they set the nominal wage rate for their particular sector. But in the short run the expected general wage level W^e and the expected price level P^e may deviate from the actual levels. For the purpose of short-run analysis, if B^u is the nominal rate of unemployment benefit, the wage-setting curve (6) should therefore be rewritten as

$$\begin{aligned} \frac{W_i}{P^e} &= m^w \left[(1-u) \frac{W^e}{P^e} + u \frac{B^u}{P^e} \right] \Leftrightarrow \frac{W_i}{P} \frac{P}{P^e} = m^w \left[(1-u) \frac{W^e}{W} \frac{W}{P} \frac{P}{P^e} + u \frac{B^u}{P} \frac{P}{P^e} \right] \Leftrightarrow \\ \frac{W_i}{P} &= m^w \left[(1-u) \frac{W^e}{W} \frac{W}{P} + ub \right], \end{aligned} \quad (20)$$

where we have used the definition of the real unemployment benefit, $b \equiv B^u/P$. The price-setting equation (3) and the assumption that mark-ups are identical across sectors imply that $W_i/P = W/P = 1/m^p$. Inserting this into (20) along with the assumption that $b = c \cdot (W/P)$, we get:

$$1 = m^w \left[(1-u) \frac{W^e}{W} + uc \right]. \quad (21)$$

In a long-run equilibrium expectations must be fulfilled ($W^e = W$), so from (21) the natural unemployment rate must satisfy:

$$1 = m^w(1 - \bar{u} + \bar{u}c), \quad (22)$$

which is equivalent to (19). Subtracting (22) from (21) and rearranging, we find:

$$u - \hat{u} = \left(\frac{1-u}{1-c} \right) \left(\frac{W^e - W}{W} \right). \quad (23)$$

This shows how a misperception of the general wage level ($W^e \neq W$) can cause unemployment to deviate from its natural rate. Indeed, we see that such expectational errors are a *necessary* condition for deviations from long-run equilibrium. However, when shocks to the economy generate expectational errors, the existence of nominal rigidities will greatly *amplify* the resulting fluctuations of employment around the natural rate, as we shall explain in detail in Chapter 17.

1.7 Long-run versus short-run economic policies and some second thoughts on the traditional dichotomy of macroeconomics

Structural policy and stabilization policy

A main motivation for studying economic phenomena is the need to improve the basis for economic policy advice. For instance, we want to understand the sources of economic growth because this could be helpful in designing a growth-promoting policy for a low-income country trying to escape poverty.

The division of macroeconomics described above leads to a parallel division of economic policies into *long-run policies* aimed at promoting growth and long-run prosperity and at reducing long-run unemployment, and *short-run policies* aimed at mitigating economic fluctuations and their harmful consequences coming, for example, from sudden increases in unemployment.

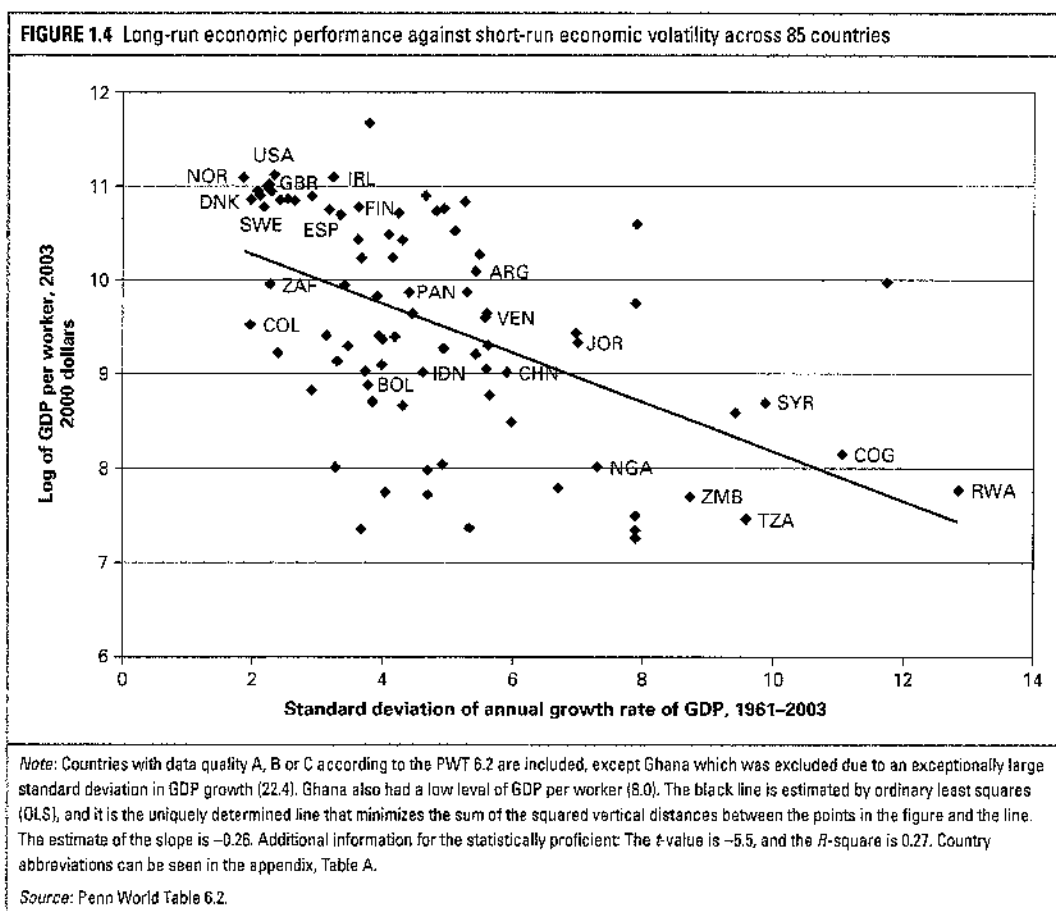
However, the division of macroeconomics suggests more than this categorization of policies according to their aims. The basic and different assumptions underlying the two parts of macroeconomic theory have consequences for the channels through which long-run and short-run policies can affect the economy.

For example, recall that long-run macroeconomics assumes full nominal wage and price adjustment. As we saw from Eq. (10), this implies that long-run unemployment is determined exclusively by the parameters m^w , m^p and b which reflect the *structural* characteristics of labour and product markets. Because it is rooted in the basic structural features of the economy, the natural rate of unemployment is also referred to as the 'structural' unemployment rate. It follows that a policy intended to reduce long-run unemployment can only be successful if it affects the economy's basic structures. Specifically, our analysis suggests that such a policy must try to increase the degree of competition in labour or product markets (by lowering the values of m^w and m^p) and/or reduce the generosity of the system of unemployment insurance (lowering b).¹³

In a similar way, policies aimed at promoting long-run growth and prosperity must affect one or several structural characteristics of the economy such as the long-run propensities to save and invest, to engage in education and R&D, etc. In short, policies for the long run must be *structural* policies.

A short-run policy to mitigate business cycles, on the other hand, can be a monetary or fiscal policy of *demand management*. Such a policy can affect the rate of employment in the short run even if it does not influence the basic structures and incentives in the economy. The reason is that in the short run prices and expectations are not fully adjusted,

13. In richer models of the labour market there are other ways of reducing the natural unemployment rate, such as improving the level and composition of work skills through education and training to attain a better match between the skills possessed by workers and the skills demanded by employers.



and hence changes in nominal demand can affect real economic variables, as we explained earlier.

Discussing the relevance of the separation of macroeconomics in two branches

In our discussion of macroeconomics so far we have taken an orthodox view: macroeconomics for the long and the short run are almost distinct fields intended to explain separate phenomena (growth and business cycles, respectively) and using conflicting assumptions and qualitatively different models. From this dichotomy it follows that long-run structural policies and short-run stabilization policies are also considered as distinct. This strict dichotomy can be and has been challenged.

Consider Fig. 1.4. Each dot represents a country. Along the horizontal axis the standard deviation of the annual growth rate of the country's GDP between 1960 and 2003 is depicted. This is a measure of the volatility of the country's economic activity, that is, of the amplitude of the business cycle. Up along the vertical axis is the log of the country's GDP per worker (that is, per member of the labour force) in 2003. This is a measure of the country's long-run economic performance, i.e., its capability to create income for its inhabitants.¹⁴ The figure includes a large number of countries for which

14. The figure and our discussion are inspired by the article by Valerie and Garey Ramey, 'Cross-Country Evidence on the Link Between Volatility and Growth', *American Economic Review*, 85, 1995, pp. 1138–1151, and by subsequent research.

data of good quality were available from the relevant source, so the sample should be quite representative.

In short, the figure plots the long-run income-creating capability of the economy against its short-run volatility in economic activity across a large number of countries. The figure shows a relatively tight negative relationship between these two measures, as indicated by the downward-sloping regression line.

You will be warned repeatedly through this textbook about the importance of not confusing correlation with causality. It cannot be concluded from the figure alone that economic volatility affects long-run economic performance negatively. This is just one out of several alternative causal relationships that could underlie the correlation of Fig. 1.4. However, several of these causal interpretations would challenge the traditional dichotomy of macroeconomics.

First, the correlation could indeed reflect that short-run volatility affects the long-run economic performance negatively. This could be because instability creates an unfavourable environment for investment in capital and education. In this case an economic model that contributes to an understanding of business cycles would also contribute to an understanding of the economy's long-run performance. And a policy that affects short-run volatility would have an impact on the economy's long-run performance. Note that this could be an important motivation for stabilization policy. Stability could be a goal, not only because of the direct costliness of instability itself, but also because of the indirect positive effect of stability on the economy's long-run performance. Of course, no economist would claim that the degree of instability of the business cycle is the only factor that explains long-run performance. The article by Ramey and Ramey referred to in Footnote 14 finds that if one controls for (includes) other traditional explaining factors such as, e.g., the average share of investment in GDP, then one finds an even stronger connection between volatility and long-run performance.

Second, the correlation of Fig. 1.4 could be due to *reversed causality*, so that it is really a good long-run performance that causes a low volatility. It could be, for instance, that rich people and highly productive firms put more emphasis on, or have better possibilities for, creating relatively stable evolutions of consumption and investment over time. In that case, an economic model that contributes to an explanation of the economy's long-run capacity to create income would also deliver part of an explanation of the severity of short-run business cycles. This challenges once again the traditional dichotomy.

Finally, the tight relationship could be an expression of *spurious correlation*, where there is no causal link between the two measures of the figure, but a third factor, say the skilfulness of a country's inhabitants or the quality of its institutions, that explains both long-run and short-run performance. Different countries are skilful to different degrees. Going across countries the more skilful ones will exhibit more stable output *and* higher income per worker, both caused by the skilfulness. In this way a correlation such as that in Fig. 1.4 can arise without any of the variables along the axes affecting each other. If this is the true explanation, it would be all right to have separate models for the short and long runs, respectively, but the third factor mentioned should then be an explanatory factor in both types of models.

Despite these reservations, we will continue along a traditional line with a rather strict dichotomy between short-run and long-run analysis. We do, however, draw an important lesson from our discussion: although Fig. 1.4 is not a proof in itself, we think that there is more thorough evidence that relative economic stability in the short run causes relatively good economic performance in the long run. Hence, an important motivation for short-run stabilization policy is that success with such policy can have desirable long-run effects on income per capita.

1.8 Summary

This chapter has explained the ideas behind the structuring of our course in macroeconomics. Understanding these ideas is important for a good understanding of this book. Let us therefore summarize our main concepts and arguments, taking the key elements in a different sequence which will help to clarify the distinction between macroeconomics for the long run and macroeconomics for the short run:

1. The economy's *long-run equilibrium* is the combination of relative prices and quantities which would emerge in a general equilibrium where wages and prices have had time to adjust fully to past shocks and where no further shocks have occurred over a sufficiently long period.
2. If the long-run equilibrium is the outcome of perfect competition, all economic agents have taken wages and prices as given and found their optimal price-taking supplies and demands, and prices have adjusted to equate these supplies and demands market by market.
3. In practice, empirical studies often find that prices are above marginal costs, indicating that most markets – including the labour market – are characterized by *imperfect competition*. Still we can think of long-run relative prices as being determined by an underlying general equilibrium system, although not one that ensures equality between price-taking supplies and demands.
4. The *natural rate* is the rate of resource utilization emerging when relative prices have fully adjusted to their long-run equilibrium values. Imperfect competition typically means that the natural rate is less than 100 per cent. When this is the case, we say that *real rigidities* prevail. Real rigidities imply that individual agents do not wish to reduce their real (relative) wages and prices very much in response to unemployment or excess capacity. Hence real wages and prices do not adjust sufficiently to prevent permanent underutilization of resources.
5. *Macroeconomics for the long run* aims at explaining the trends in main economic time series and the effects of structural economic policies. In long-run macroeconomics the economy is analysed as if relative prices are fully adjusted to their long-run equilibrium values in each period, the fundamentals of the economy such as preferences and technology evolve smoothly and predictably, and expectations are correct all the time. One implication is that in long-run macroeconomics aggregate output is determined from the supply side alone, as the level of output that can be produced when available resources are utilized at their natural rates.
6. *Short-run nominal wage and price rigidities* mean that some money wages and/or prices are fixed over a certain period. Empirical evidence shows that most money prices and money wages are only adjusted with certain time intervals even under considerable inflation. Thus nominal rigidities prevail in the short run, and these may cause the rate of resource utilization to deviate from the natural rate for periods of sufficient length to be of interest.
7. Individual agents adjust their nominal wages or prices with the purpose of changing their real wages or prices. Because real rigidities imply that agents do not want to change their real prices very much in response to changes in economic activity, real rigidities also tend to generate nominal rigidities. When the degree of real rigidity is strong, short-run nominal wage and price rigidities can be privately optimal even if the *menu costs* of nominal wage and price adjustment are very small. At the same time the social cost of nominal rigidity can be many times as large as the perceived private cost for the individual agent.

8. Short-run nominal wage and price rigidities may imply that some *relative* prices are also fixed in the short run. For instance, if both nominal wages and nominal prices are fixed, real wages are fixed. Short-run nominal rigidities can explain why it takes time for relative (real) prices to adjust to their long-run equilibrium levels.
9. *Macroeconomics for the short run* seeks to explain the fluctuations in main economic time series around their trends and the effects of stabilization policies. Economists believe that exogenous shocks (sudden unpredictable changes in factors such as business confidence, preferences and technology), short-run nominal wage and price rigidities, and expectational errors are fundamental for understanding short-run fluctuations. Because of nominal rigidities and expectational errors, the actual rate of resource utilization can deviate from the natural rate. In the short run aggregate demand is therefore just as important for economic activity as aggregate supply.
10. International evidence shows a clear positive correlation between economic stability in the short run and economic performance (income creation per capita) in the long run. Various causal relationships can explain this pattern, but one plausible explanation is that stability affects long-run performance positively. This constitutes an important motivation for stabilization policy.

1.9 Exercises

Exercise 1. Aggregation in macroeconomics

Discuss why macroeconomists typically work with aggregate variables and explain the arguments they use to defend this procedure. Do you find the arguments reasonable? Justify your answers.

Exercise 2. Short-run versus long-run macroeconomics

Explain the different assumptions underlying macroeconomics for the short run and macroeconomics for the long run. Do you find these assumptions reasonable? Justify your answers.

Exercise 3. Macroeconomic policies for the short run and for the long run

Explain the difference between structural policies and demand management policies. Explain the concepts of 'nominal rigidities', 'real rigidities' and the 'natural rate of resource utilization'. Try to give concrete examples of typical short-run policies and typical long-run policies.

Exercise 4. Fighting structural unemployment

Try to think of specific policies which might reduce the natural rate of unemployment. Explain why you think that these particular policies might reduce long-run unemployment.

Exercise 5. Menu costs and real and nominal rigidities

Explain and discuss the type of costs included in the concept of 'menu costs'. Explain why and in what sense so-called real rigidities are necessary to generate nominal wage and price rigidity.

Book One

**The Long Run
Economic Growth,
Long-run
Unemployment and
Structural Economic
Policy**

